# Building The Void

## BBP as the Unique Survivor of Three Structural Prohibitions

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## Abstract

The Bailey–Borwein–Plouffe formula was found by PSLQ, an integer-relation algorithm that does not search for formulas. PSLQ defines a null space and returns whatever survives inside it. This paper takes one step back from the formula and reconstructs the void that PSLQ was searching — the set of structural prohibitions for which BBP is the unique residue.

Six results are established by direct computation. First, the BBP numerator is not a weighting of the eight shell channels; it is a pole annihilator. Written over the integral representation, the numerator shares two quadratic factors with the denominator and cancels six of its eight poles, leaving exactly one real pair and one complex pair. Second, base 16 and shell spacing 8 are not two independent facts: both are read off the single surviving complex pole at $\sqrt{2}\cdot e^{(i\pi/4)}$. The pole argument of 45° forces period 8; the pole modulus $\sqrt{2}$ raised to that period forces base 16. Third, the subspace of numerators whose integral is a rational multiple of π has dimension exactly 2, verified by exact rank computation and confirmed against a 250-digit integer-relation search. Fourth, that two-dimensional void contains a one-dimensional zero-relation line. Fifth, the DC-blocking condition — the requirement that the coefficients sum to zero — cuts the void to a unique line, excludes the zero relation, and that line is spanned by BBP. Sixth, in the reduced two-arm form the three surviving conditions correspond to circuit facts that are structural rather than parametric: a wire absent, a differential split, and a 2:1 ratio.

Two claims made earlier in this thread are retracted here with their falsifying computations. The boundary-zero condition $N(1)=0$ was claimed to cause the logarithmic cancellation; it is neither necessary nor sufficient, and counterexamples of both kinds are given. The two-arm circuit form was claimed to be more accurate than the four-channel form; Monte Carlo shows it is measurably worse in error magnitude. What survives is a different and stronger statement: the two-arm differential topology confines the output to the one-dimensional π-line of transcendental space under arbitrary component drift, reducing error dimensionality from seven to one.

## 1. One Step Behind PSLQ

PSLQ did not look for a formula for π. It was handed a vector of real numbers and asked a single question: is there a small integer combination of these that vanishes? The BBP formula is what came back. The formula was never the target; it was the residue of a constraint.

This is the standard shape of the method used throughout this program. Do not chase the named object. Ask what prohibitions make it unavoidable. The route is the content; the object is what is left over when the route is walked. Applied here it produces a specific question:

> *What is the void that BBP is the only thing living in?*

The answer requires leaving the series form. In series form the formula presents itself as four weighted channels — coefficients {+4, −2, −1, −1} attached to denominators 8k+1, 8k+4, 8k+5, 8k+6 — and every structural question becomes a question about those four weights. This paper shows that the four channels are not the object. They are a projection of a two-pole object, and almost everything that appears mysterious in the weights becomes forced once the object is seen at its own rank.

### 1.1 The integral representation

Expanding the geometric factor 16^(−k) as an integral converts each shell sum into a definite integral over the unit interval. For denominators of the form 8k+r:

$$S(r) = \Sigma[k\geq 0]\ 1 / (16^k\,(8k+r)) = 16 \int_0^1 t^{(r-1)} / (16 - t^8)\, dt$$

Verified to 50 digits for $r \in \{1,4,5,6\}$. Assembling the BBP combination with numerator exponent $p = r - 1$:

$$\pi = 16 \int_0^1 ( 4 - 2t^3 - t^4 - t^5 ) / ( 16 - t^8 )\, dt$$

S(1): series = 1.007184476415  integral = 1.007184476415  match = True

S(4): series = 0.255412811883  integral = 0.255412811883  match = True

S(5): series = 0.205002557636  integral = 0.205002557636  match = True

S(6): series = 0.171317070666  integral = 0.171317070666  match = True

BBP series   = 3.141592653589793

BBP integral = 3.141592653589793

pi        = 3.141592653589793  match: True

Every quantity in this integrand is algebraic. The numerator is a degree-5 polynomial with integer coefficients; the denominator is a degree-8 polynomial with integer coefficients; the integration limits are 0 and 1. There is no transcendental content anywhere in the parts list. π is what the assembly produces, not what it contains. This is an engineering constraint, not a philosophical remark: the machine has no storage slot for the constant it outputs.

## 2. Result 1 — The Numerator Is a Pole Annihilator

The denominator factors completely over the rationals into four quadratics:

$$16 - t^8 = (2 - t^2)(2 + t^2)(t^2 - 2t + 2)(t^2 + 2t + 2)$$

The numerator also factors, and this is the point at which the structure opens:

$$4 - 2t^3 - t^4 - t^5 = -(t - 1)(t^2 + 2)(t^2 + 2t + 2)$$

Two of the three numerator factors are denominator factors. $(t^2 + 2)$ is the same polynomial as $(2 + t^2)$. $(t^2 + 2t + 2)$ appears verbatim on both sides. The rational function therefore reduces:

$$N(t)/D(t) = (t - 1) / [ (t^2 - 2)(t^2 - 2t + 2) ]$$

```
16 - t^8 factors  : -(t**2 - 2)*(t**2 + 2)*(t**2 - 2*t + 2)*(t**2 + 2*t + 2)
numerator factors : -(t - 1)*(t**2 + 2)*(t**2 + 2*t + 2)
reduced form      : (t - 1)/(t**4 - 2*t**3 + 4*t - 4)
identity holds    : True
```

The BBP integrand is not a degree-5-over-degree-8 rational function. It is degree-1-over-degree-4. Six of the eight poles are cancelled outright by the numerator.

### 2.1 Which poles survive

All eight roots of $16 - t^8$ lie on the circle of radius $\sqrt{2}$, spaced 45° apart. Listing them by angle:

| Pole | Angle | Modulus | From factor | Status |
|---|---|---|---|---|
| $+\sqrt{2}$ | 0° | 1.414214 | $t^2 - 2$ | SURVIVES |
| $1 + i$ | 45° | 1.414214 | $t^2 - 2t + 2$ | SURVIVES |
| $+i\sqrt{2}$ | 90° | 1.414214 | $t^2 + 2$ | killed |
| $-1 + i$ | 135° | 1.414214 | $t^2 + 2t + 2$ | killed |
| $-\sqrt{2}$ | 180° | 1.414214 | $t^2 - 2$ | SURVIVES |
| $-1 - i$ | 225° | 1.414214 | $t^2 + 2t + 2$ | killed |
| $-i\sqrt{2}$ | 270° | 1.414214 | $t^2 + 2$ | killed |
| $1 - i$ | 315° | 1.414214 | $t^2 - 2t + 2$ | SURVIVES |

The survivors are exactly the 0°/180° axis and the ±45° diagonal. The numerator kills the 90° axis and the 135° diagonal.

> *This is the same 45° selection that the earlier forced chain arrived at from the integration-limit argument, now recovered from a different direction. There it came from requiring cos(45°) = 2^(−1/2) to be a mark*

> *coordinate of the frame. Here it comes from asking which poles the numerator does not cancel. Two traversals, one location.*

The surviving set is also minimal for the job. Exactly one real pair and one complex pair remain. A real pole contributes a logarithm on integration; a complex pair contributes both a logarithm and an arctangent. To cancel a logarithm you need a second logarithm of the opposite sign, so at least one real pair and one complex pair must be present. Any smaller surviving set cannot produce a cancellation; the one selected here is the minimum that can.

### 2.2 The two arms

Partial fractions on the reduced form give the object at its own rank:

$$(t - 1)/[(t^2 - 2)(t^2 - 2t + 2)] = t/[4(t^2 - 2)] - (t - 2)/[4(t^2 - 2t + 2)]$$

Two arms. Integrating each over [0,1]:

| Arm | Poles | Type | $\int_0^1$ |
|---|---|---|---|
| A: $t / 4(t^2-2)$ | $\pm\sqrt{2}$ | real | $-\log(2)/8$ |
| B: $-(t-2) / 4(t^2-2t+2)$ | $1 \pm i$ | complex | $+\log(2)/8 + \pi/16$ |
| Sum | — | — | $\pi/16$ |

The logarithms annihilate exactly. The arctangent has no partner and survives alone. That survivor is π/16, and multiplying by the 16 in the integral representation gives π.

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## 3. Result 2 — Base 16 and Shell 8 Are One Fact

The surviving complex pole sits at $1 + i = \sqrt{2}\cdot e^{i\pi/4}$. Expanding its factor as a power series gives a closed form for every coefficient:

$$1/(t^2 - 2t + 2) = \Sigma[j\geq 0]\ t^j \cdot \sin((j+1)\pi/4) / 2^{(j+1)/2}$$

Verified coefficient-by-coefficient against the symbolic series expansion through j = 15.

```
j   series coeff   sin((j+1)pi/4)/2^((j+1)/2)  match
0        1/2                 1/2   True
1        1/2                 1/2   True
2        1/4                 1/4   True
3         0                  0   True
4        -1/8                -1/8   True
5        -1/8                -1/8   True
6        -1/16               -1/16   True
```

| | | | |
|---|---|---|---|
| 7 | 0 | 0 | True |
| 8 | 1/32 | 1/32 | True |

... verified through $j = 15$ : True

Two constants fall out of this one expression, and they are not independent.

**The shell is the pole argument**

The phase factor is $\sin((j+1)\cdot 45°)$. Advancing 45° per step returns to the starting phase after exactly 8 steps. The shell spacing 8 is the period of the pole's argument, nothing more.

**The base is the pole modulus raised to the shell**

The amplitude factor is $2^{(-(j+1)/2)}$, a decay of $\sqrt{2}$ per step. Over one full shell of 8 steps the accumulated decay is $(\sqrt{2})^8 = 16$. The base is the modulus raised to the period.

> *Base 16 and shell 8 are not two design choices that happen to fit. They are the modulus and the argument of a single pole at $\sqrt{2}\cdot e^{(i\pi/4)}$, read in two different ways. One location, two readings.*

### 3.1 The blind positions

The phase factor vanishes where $\sin((j+1)\pi/4) = 0$, at $j = 3$ and $j = 7$, corresponding to shell positions $r = p+1 = 4$ and $r = 8$. These are the 180° and 360° marks. A pole at 45° cannot address the directions orthogonal to its own axis pair; those two positions of every shell are structurally unreachable from the complex arm alone.

Note that $r = 4$ appears in BBP with coefficient −2 despite the complex arm being blind there. That contribution comes entirely from the real arm. The division of labour is visible in the reduced form: the complex arm carries the angular content and cannot reach $r \in \{4,8\}$; the real arm covers what the complex arm cannot, and is then cancelled away in the logarithmic channel.

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## 4. Result 3 — The Void Has Dimension Two

Fix the denominator $16 - t^8$ and let the numerator range over all polynomials of degree at most 7 — an 8-dimensional space, one dimension per shell position. Define the void as the set of numerators whose integral is a rational multiple of π. This is a statement about the value, not about any chosen decomposition, so it is basis-independent.

### 4.1 Exact computation

Solving the general partial-fraction system symbolically gives each of the four arm pairs as an explicit linear function of the numerator coefficients $n_0 \ldots n_7$:

A1 = -n1/32 - n3/16 - n5/8 - n7/4    B1 = -n0/32 - n2/16 - n4/8 - n6/4

A2 =  n1/32 - n3/16 + n5/8 - n7/4    B2 =  n0/32 - n2/16 + n4/8 - n6/4

A3 = -n0/64 + n2/32 + n3/16 + n4/16  B3 =  n0/32 + n1/32 - n3/16 - n4/8

  - n6/8 - n7/4                - n5/8 + n7/4

```
A4 =  n0/64 - n2/32 + n3/16 - n4/16   B4 =  n0/32 - n1/32 + n3/16 - n4/8
   + n6/8 - n7/4                         + n5/8 - n7/4
```

Integrating each arm over [0,1] and collecting the six non-π transcendentals — log 2, log(1+√2), log 3, log 5, arctan(1/√2), arctan 2 — gives a 6 × 8 coefficient matrix:

```
       log 2 : n7/2
  log(1+sqrt2) : sqrt(2)*(n0 + 2*n2 + 4*n4 + 8*n6)/64
       log 3 : n1/64 - n3/32 + n5/16 - n7/8
       log 5 : n0/128 - n2/64 + n3/32 - n4/32 + n6/16 - n7/8
arctan(1/sqrt2): sqrt(2)*(n0 - 2*n2 + 4*n4 - 8*n6)/64
    arctan 2 : n0/64 - n1/32 + n2/32 - n4/16 + n5/8 - n6/8


Rank = 6      LOG-FREE SUBSPACE DIMENSION = 2
```

Rank 6 out of 8 columns. The void is 2-dimensional. A basis, cleared to integers:

| Vector | $n_0 \ldots n_7$ | $16\cdot\int$ | $\Sigma\, n_p$ |
|---|---|---|---|
| $v_1$ | [−4, 0, 0, 2, 1, 1, 0, 0] | −π | 0 |
| $v_2$ | [0, −8, −4, −4, 0, 0, 1, 0] | −2π | −15 |

$v_1$ is the BBP formula up to sign. Both were verified by direct series summation to 30 significant digits and by 250-digit numerical integration.

```
v1 (= -BBP):  n = [-4, 0, 0, 2, 1, 1, 0, 0]
   series sum = -3.14159265358979323846264338328
   expected   = -3.14159265358979323846264338328   match = True


v2:         n = [0, -8, -4, -4, 0, 0, 1, 0]
   series sum = -6.28318530717958647692528676656
   expected   = -6.28318530717958647692528676656   match = True
```

## 4.2 Independence of the transcendental basis

The rank-6 count is only meaningful if the six non-π constants are rationally independent of each other and of π. An initial PSLQ run at 90 digits returned an apparent relation with coefficients near $4\times10^{11}$ and a residual of $3.6\times10^{-70}$. This was recorded, then tested at higher precision.

| Precision | Result |
|---|---|
| 90 digits | relation found, max coefficient 3.938e+11 |
| 200 digits | NO relation up to coefficient bound $10^{12}$ |

| Precision | Result |
|---|---|
| 400 digits | NO relation up to coefficient bound $10^{12}$ |

Seven constants with coefficients bounded by $10^{12}$ require roughly 84 digits of working precision. At 90 digits the search was operating with six digits of margin, which is not enough. The relation disappears when the margin is widened, which is the signature of a PSLQ artifact rather than a real dependency. Logged as a caught artifact; the rank-6 count stands.

### 4.3 Independent confirmation of the dimension

As a cross-check that does not use the symbolic rank at all, the integer-relation lattice of [I(0) … I(7), π] was searched directly at 250 digits, where $I(p) = \int_0^1 t^p/(16-t^8)\,dt$. Both known relations verify to residual 0. A generic sample of vectors outside span$\{v_1, v_2\}$ was tested and none produced a rational multiple of π; every tested element of the span did.

```
n = [1, 0, 0, 0, 0, 0, 0, 0]    -> NOT a rational multiple of pi
n = [0, 1, 0, 0, 0, 0, 0, 0]    -> NOT a rational multiple of pi
n = [1, 1, 1, 1, 1, 1, 1, 1]    -> NOT a rational multiple of pi
n = [4, -2, -1, -1, 0, 0, 0, 0]  -> NOT a rational multiple of pi
n = [3, 0, 0, -1, 0, 0, 0, 0]   -> NOT a rational multiple of pi


1*v1 + 0*v2  -> = -1/16 * pi
0*v1 + 1*v2  -> = -1/8  * pi
1*v1 + 1*v2  -> = -3/16 * pi
2*v1 - 3*v2  -> = +1/4  * pi
5*v1 + 7*v2  -> = -19/16 * pi
```

Note the fourth line of the first block. The naive coefficient pattern [4, −2, −1, −1] placed on consecutive exponents p = 0,1,2,3 is not in the void. The BBP weights are not intrinsically special as a number pattern; they are special only when attached to the exponents 0, 3, 4, 5.

---

## 5. Result 4 — The Zero-Relation Line

Inside the two-dimensional void sits a distinguished line: the numerators whose integral is not merely a rational multiple of π but exactly zero. It is spanned by $2v_1 - v_2$:

$$z = [\,-8, 8, 4, 8, 2, 2, -1, 0\,]$$

$$\sum_{k\geq 0} 16^{-k} \left[ -\frac{8}{8k+1} + \frac{8}{8k+2} + \frac{4}{8k+3} + \frac{8}{8k+4} + \frac{2}{8k+5} + \frac{2}{8k+6} - \frac{1}{8k+7} \right] = 0$$

```
z = 2*v1 - v2 = [-8, 8, 4, 8, 2, 2, -1, 0]
sum z_p I(p) = 2.61882e-251    exactly zero at 250 dps: True
```

```
coefficient sum(z_p) = 15
```

This is the base-16 zero relation — the object whose dimension is the subject of the open problem RP2 of Borwein, Borwein and Galway (2004), pursued elsewhere in this program. It appears here not as a separate investigation but as a sub-object of the void: the kernel of the map from void to Qπ.

Its presence matters for the selection argument in the next section. A search that only demanded 'rational multiple of π' would happily return the zero relation, which carries no information about π at all. Any correct account of what singles out BBP must exclude it.

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## 6. Result 5 — DC-Blocking Selects BBP Uniquely

The void is two-dimensional, so 'lives in the void' does not by itself pick out BBP. One further condition is needed, and it is the one already visible in the formula and previously noted in this program as unremarked in the literature: the coefficients sum to exactly zero.

$$+4 - 2 - 1 - 1 = 0$$

For a general void element $a \cdot v_1 + b \cdot v_2$, the coefficient sum is a linear functional:

$$\Sigma\, n_p = a \cdot 0 + b \cdot (-15) = -15b$$

It vanishes if and only if $b = 0$. The DC-blocked subspace of the void is therefore exactly the line spanned by $v_1$, and $v_1$ is BBP.

| Element | Coefficient sum | Value (×π) |
|---|---|---|
| $1 \cdot v_1 + 0 \cdot v_2$ | 0 | −1 |
| $0 \cdot v_1 + 1 \cdot v_2$ | −15 | −2 |
| $2 \cdot v_1 - 1 \cdot v_2$ | 15 | 0 (zero relation) |
| $1 \cdot v_1 + 1 \cdot v_2$ | −15 | −3 |
| $3 \cdot v_1 + 2 \cdot v_2$ | −30 | −7 |

The condition does two jobs at once. It excludes the zero relation, whose coefficient sum is 15. And it cuts the remaining two-dimensional space to a unique line.

### 6.1 The complete void

Three prohibitions, one survivor:

```
8-dimensional numerator space   (deg N <= 7)


 [P1] proper rational function -- shell structure 8k+r exists
```

```
[P2] value must be a rational multiple of pi
  => LOG-FREE SUBSPACE              dim 2
    spanned by v1 -> -pi/16,  v2 -> -pi/8

  contains the zero-relation line        dim 1
    2*v1 - v2, coefficient sum 15

[P3] coefficients must sum to zero (DC-blocking)
  => UNIQUE LINE                 dim 1
    v1 alone  ==  BBP
```

> *Nothing was constructed. Three things were forbidden and one shape stood up. No space, no metric, no search over formulas — only prohibitions, and the residue.*

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## 7. Correction Log

Two claims made earlier in this thread, before these computations were run, are retracted here. Both are recorded with the computation that killed them, per the standing protocol that a killed claim is a constraint pointing somewhere and is therefore worth more than a quiet deletion.

### ⊥ Correction 1 — The boundary-zero mechanism

CLAIMED: the factor $(t - 1)$ in the numerator is what couples the two arms and forces the logarithms to cancel; the boundary zero is the mechanism of the cancellation.

STATUS: FALSE. Neither necessary nor sufficient.

Working in the reduced form with numerator $M(t) = (at+b)(t^2-2t+2) + (ct+d)(t^2-2)$, the integral is:

$$16\cdot\int_0^1 = 4\pi(c+d) - 8(a+c)\cdot\log 2 - (16b/\sqrt{2})\cdot\log(1+\sqrt{2})$$

The log-free condition is therefore two conditions, $b = 0$ and $a + c = 0$ — a codimension-2 condition. The boundary-zero condition is $M(1) = a + b - c - d = 0$, a codimension-1 condition. Different objects entirely.

| Test | (a,b,c,d) | M(t) | M(1) | Integral | Verdict |
|---|---|---|---|---|---|
| A | (1,0,−1,0) | $-2t^2+4t$ | 2 (≠0) | $-\pi/4$ (log-free) | not necessary |
| B | (1,1,1,1) | $2t^3-2t$ | 0 | contains log | not sufficient |

WHAT SURVIVES: the boundary zero is not the mechanism of cancellation, but it is the selector. Within the void, $N(1) = \Sigma\, n_p$, so the boundary zero and the DC-blocking condition are the same condition, and it is what cuts the void from dimension 2 to dimension 1. The role was misidentified, not the importance. Corrected statement in Section 6.

### ⊥ Correction 2 — The accuracy claim

CLAIMED: because the two-arm form enforces its balance topologically rather than by resistor matching, it dissolves the coefficient-precision wall identified in the prior analog read-head work.

STATUS: FALSE as stated. Monte Carlo, 20,000 trials, 1% independent component tolerance:

| Model | median \|err\| | p99 \|err\| | > 1 hex bin |
|---|---|---|---|
| four independent weights {4,−2,−1,−1} | 2.763e−02 | 1.034e−01 | 12.50% |
| two arms, four independent params | 4.820e−02 | 1.846e−01 | 38.12% |
| two arms, differential drive | 4.723e−02 | 1.797e−01 | 36.97% |

The two-arm form is worse in error magnitude by roughly a factor of 1.7 at the median. The precision wall is not dissolved.

WHAT SURVIVES: a different property, established in Section 8, which is about the dimensionality of the error rather than its size.

### △ Caught artifact — spurious PSLQ relation

A 90-digit PSLQ run reported an integer relation among the seven transcendentals with coefficients near $4\times10^{11}$. Recorded, then retested at 200 and 400 digits, where it vanished. Seven constants at coefficient bound $10^{12}$ need about 84 digits; 90 left six digits of margin. Artifact, not dependency. The rank-6 result was independently confirmed by the direct lattice search in Section 4.3, which does not depend on the transcendental basis at all.

### △ Caught artifact — a basis-dependent reading

During the coefficient analysis the π-coefficient map appeared as $n_1/64 - n_5/16$, which reads as: π enters BBP only through the 8k+6 channel, with the other three channels contributing none. This looked like a structural finding and is not one.

```
arctan(2) + arctan(1/2) = pi/2  :  True


I(5) = int_0^1 t^5/(16-t^8) dt = 0.01070731691665609118542O8
 basis A (arctan 2)  : -pi/16 + arctan(2)/8 + log3/16   -> pi-coeff = -1/16
 basis B (arctan 1/2):       -arctan(1/2)/8 + log3/16  -> pi-coeff =  0
 both equal I(5) : True
```

arctan 2 already contains π/2. Choosing it as a basis element hides π inside it; choosing arctan(1/2) instead moves the π back out. The split of a value into a π part and an arctangent part is a property of the reader's basis, not of the formula. This is precisely a reader-side artifact treated as an object-side property, and it is retracted.

Every result in Sections 2 through 6 was re-examined against this. The pole-annihilation identity is a polynomial identity with no basis involved. The void dimension is defined by membership in Qπ, a property of the value. The DC-blocking selection is a condition on integer coefficients. The shell and base derivation is a statement about a pole location. All survive; the channel-attribution claim does not.

## 8. Result 6 — Error Dimensionality, Not Error Magnitude

The corrected engineering claim is sharper than the one it replaces. It concerns where the error lives, not how big it is.

### 8.1 The three conditions as circuit facts

In the reduced two-arm form the surviving structure is:

$$16\cdot\int_0^1 = 4\pi(c+d) - 8(a+c)\cdot\log 2 - (16b/\sqrt{2})\cdot\log(1+\sqrt{2})$$

Each condition has a realization that is not a trimmed value:

| Condition | Algebra | Circuit realization | Exact? |
|---|---|---|---|
| kill $\log(1+\sqrt{2})$ | $b = 0$ | no DC path wired into the real arm | yes — an absence |
| kill log 2 | $a + c = 0$ | one source, antisymmetric split (differential) | yes — topology |
| DC-block, $N(1)=0$ | $d = 2a$ | 2:1 ratio (center tap) | yes — power of two |

With all three held, the residual output is $4\pi(c+d) = 4\pi(-a + 2a) = 4\pi a$. A single 4:1 divider setting $a = 1/4$ yields π.

### 8.2 Behaviour under drift

Perturbing a while the three structural conditions hold — which they do automatically, since none of them is a component value:

| a (drifted) | pi coeff | log2 coeff | logsqrt coeff |
|---|---|---|---|
| 0.250000 | 1.000000000000 | -0.000e+00 | -0.000e+00 |
| 0.250250 | 1.001000000000 | -0.000e+00 | -0.000e+00 |
| 0.252500 | 1.010000000000 | -0.000e+00 | -0.000e+00 |
| 0.262500 | 1.050000000000 | -0.000e+00 | -0.000e+00 |
| 0.300000 | 1.200000000000 | -0.000e+00 | -0.000e+00 |
| 0.375000 | 1.500000000000 | -0.000e+00 | -0.000e+00 |

Under a 50% gain error the logarithmic channels remain at exactly zero. The output is still an exact rational multiple of π; only which rational it is has moved.

The four-channel form has no such confinement. A 1% error on the +4 channel alone:

```
Weights [4.04, -2, -1, -1]:
        pi: +1.000000000000
      log2: -0.000000000000
  log(1+sqrt2): +0.014142135624  <-- foreign transcendental
      log3: +0.000000000000
      log5: +0.005000000000  <-- foreign transcendental
 arctan(1/sqrt2): +0.014142135624 <-- foreign transcendental
     atan 2: +0.010000000000  <-- foreign transcendental
```

> *A single 1% weight error injects four foreign transcendentals. No gain trim removes them, because they are not multiples of π. The four-channel error lives in a seven-dimensional space; the two-arm differential error lives on a line.*

Stated plainly: the two-arm differential topology does not make the machine more accurate. It makes the machine's error one-dimensional and therefore closable by a single reference. The four-channel topology produces errors that no scalar calibration can reach.

### 8.3 What this does and does not change for the read-head

Honest scope. The reduced form gives π as a whole through a sweep-and-integrate architecture: ramp t from 0 to 1, two analog dividers, one summer, one integrator, output π/16 at end of sweep. It does not by itself perform digit extraction at position n, which still requires the modular head/tail split described in the prior analog read-head work.

What changes is the summing stage — the place where π actually appears. It collapses from four precision-matched channels to two arms whose balance conditions are wiring facts. The fill-and-spill metering, Adler injection locking, dither gate and SILR lattice anchoring developed for the ×16 modular chain are unaffected and still required.

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## 9. Status Ledger

| Result | Statement | Status |
|---|---|---|
| R1 | Numerator annihilates 6 of 8 poles; $N/D = (t-1)/((t^2-2)(t^2-2t+2))$ | Ψ verified |
| R1b | Survivors are the 0°/180° axis and the ±45° diagonal | Ψ verified |
| R2 | Shell 8 = pole argument; base 16 = pole modulus^8 | Ψ verified |

| Result | Statement | Status |
|---|---|---|
| R2b | Complex arm blind at r = 4 and r = 8 | Ψ verified |
| R3 | Log-free subspace has dimension exactly 2 | Ψ verified |
| R4 | Void contains a 1-dim zero-relation line, $2v_1 - v_2$ | Ψ verified |
| R5 | DC-blocking cuts the void to a unique line = BBP | Ψ verified |
| R6 | Two-arm differential error is 1-dimensional along π | Ψ verified |
| C1 | Boundary zero causes log cancellation | ⊥ falsified |
| C2 | Two-arm form is more accurate than four-channel | ⊥ falsified |
| A1 | 90-digit PSLQ relation among the seven constants | △ artifact |
| A2 | π enters only through the 8k+6 channel | △ artifact |
| Ω1 | Does the void construction generalize to other BBP bases? | Ω open |
| Ω2 | Physical two-arm sweep-integrate build | Ω open |
| Ω3 | Two-arm form compatible with modular digit extraction? | Ω open |

## 10. Open Problems

### Ω1 — Generality of the void construction

Every step here was carried out for base 16 with shell 8. The method is not obviously specific to that case: take a BBP-type denominator b − t^n, compute the void dimension by the same rank calculation, locate the zero-relation sub-line, and test whether DC-blocking cuts to a unique element. If the pattern holds across bases, the void construction is a general classification tool for BBP-type formulas rather than a fact about π. If it fails — if some base has a void of dimension 3, or a void where DC-blocking does not cut to a line — that failure locates exactly what is special about base 16. Both outcomes are informative. Adegoke's base-3 formulas and Bellard's mixed-moduli formula are the obvious stress tests.

### Ω2 — The two-arm sweep-integrate build

The reduced integrand is bounded and smooth on [0,1]: the real arm ranges over [−0.25, 0], the complex arm sits near 0.25 throughout with a mild bulge, and the total goes to zero at t = 1. Both denominators stay bounded away from zero on the integration path ($t^2 - 2 \in [-2,-1]$, $t^2 - 2t + 2 \in [1,2]$). Nothing rails. This should build with a ramp generator, two analog dividers, a summer and an integrator, with the three structural conditions realized as wiring. The open question is whether the measured output stays on the π-line to the precision the model predicts, or whether some second-order effect — divider nonlinearity, integrator leakage, ramp nonlinearity — reintroduces a foreign transcendental that the ideal analysis does not see.

### Ω3 — Two-arm form and digit extraction

The sharpest open question. Digit extraction at position n requires modular reduction, which is natural in the four-channel series form and not obviously available in the two-arm form. Does a two-arm modular architecture exist? The complex arm's coefficient $\sin((j+1)\pi/4)/2^{((j+1)/2)}$ is an 8-periodic phase on a $\sqrt{2}$ geometric decay, which is exactly the behaviour of a single quadrature resonator advancing 45° per step. If the modular head can be run on one resonator plus one real channel rather than four independent modular cells, the entire read-head architecture simplifies by a factor of two in channel count. Whether the modular reduction commutes with the partial-fraction decomposition is the technical question, and it is not settled here.

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## Appendix — Reproduction

All results above were produced by direct computation in a single session on 4 August 2026. Numerical work used mpmath at 50–400 decimal digits; symbolic work used SymPy exact rational arithmetic; Monte Carlo used NumPy with seed 20260804. The consolidated verification script reproduces every figure quoted in this paper.

```
R1  identity holds              : True
R2  sine-series form verified to j=15 : True
R2  blind shell positions r        : [4, 8]
R3  non-pi coefficient matrix rank    : 6 / 8
R3  LOG-FREE SUBSPACE DIMENSION      : 2
R4  v1 -> -1.0 * pi   coeff sum 0
R4  v2 -> -2.0 * pi   coeff sum -15
R4  z  -> 2.61882e-251  coeff sum 15  (exact zero at 250 dps)
R5  DC-block on a*v1+b*v2: sum = -15b  =>  unique line b=0 = BBP
R6  4-channel  median |err| 2.763e-02   p99 1.034e-01
R6  2-arm diff median |err| 4.723e-02   p99 1.797e-01
R6  2-arm log leakage max          : 0.000e+00  (exactly 0)
```

Prior work in this thread: The Address Decoder; The Angle Register; Kernel / Frame / Winding / Residue; The Read Head: BBP as the Integer's Address in the Pi-Lattice ROM; The Analog Read-Head. This paper supersedes the mechanism claims of the last two where they conflict, as recorded in Section 7.